# INNERCORE SOVEREIGN AI WORKBENCH

OFFICIAL INVESTIGATION DELIVERABLE · 100% ON-PREMISE AIR-GAPPED

## INVESTIGATION PARAMETERS

Query: Review safety and compliance evidence for cryogenic pump seals under ISO-9001.

Operator: OPERATOR_L2 | Agent: Maintenance Agent | Model: llama3.2:latest (Local)

Network Boundary: AIR-GAPPED ON-PREMISE | External Egress: ZERO (0)

## EXECUTIVE RESULT & SYNTHESIS

## GROUNDED EVIDENCE SOURCES

1. Quality_Assurance_Protocol_QA900.md · Section 1: Quality Assurance Protocol QA- (Relevance: 54.2%)
   "Calibrated sensors require recalibration every 90 days in accordance with ISO-9001 standards."
2. Plant_Safety_Standard_ISO14001.md · Section 1: Industrial Safety Standard ISO (Relevance: 53.9%)
   "# Industrial Safety Standard ISO-14001 All plant operators must adhere to strict Lockout/Tagout (LOTO) protocols prior to servi..."
3. operational_test.txt · Section 1 (Relevance: 47.7%)
   "Subsystem Alpha operates under 450 bar pressure at 1200 RPM."

## 7-STAGE AGENT EXECUTION TRACE

[TASK INTAKE] Role OPERATOR_L2 verified · Boundary: On-premise air-gapped
[MODEL ROUTER] Task: TECHNICAL_ANALYSIS · Best available local model for grounded engineering & teleme...
[AGENT PLANNING] Maintenance Agent planned investigation: 1. Hybrid semantic retrieval across private ...
[KNOWLEDGE SEARCH] Global Corpus · Retrieved 5 relevant passages from Quality_Assurance_Protocol_QA900...
[TOOL EXECUTION] Executed semantic_evidence_verifier: 5 chunks analyzed for relevance
[VERIFY] All sources & parameters verified on-premise · Zero external cloud egress
[RESULT] Inference complete (LOCAL_INFERENCE) via llama3.2:latest

## CRYPTOGRAPHIC AUDIT PROVENANCE

Execution ID: EXEC-1788827113473

SHA-256 Hash: 5863813977f2cd6e19bdb6c4a1e52d07f61a1d871d79592468cb745973c38b9d

Chain Verification: VALID (Tamper-evident mathematical hash chain)